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Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 15	
TITLE: Exception Handling using Finally, Throw, and Throws	

Problem Statement

Write a Java program to demonstrate exception handling using finally, throw, and throws.

Learning Objectives

To implement finally, throw, and throws for effective exception management.

1. **Create a Login program that throws an exception for invalid password and uses finally block.**

Programme

```java
import java.util.Scanner;

public class Login {

    static void checkPassword(String password) throws Exception {
        if (!password.equals(anObject: "12345")) {
            throw new Exception(message: "Invalid password!");
        }

        System.out.println(x: "Login successful!");
    }

    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print(s: "Enter password: ");
            String password = sc.nextLine();

            checkPassword(password);
        }
        catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
        finally {
            System.out.println(x: "Login process completed.");
        }

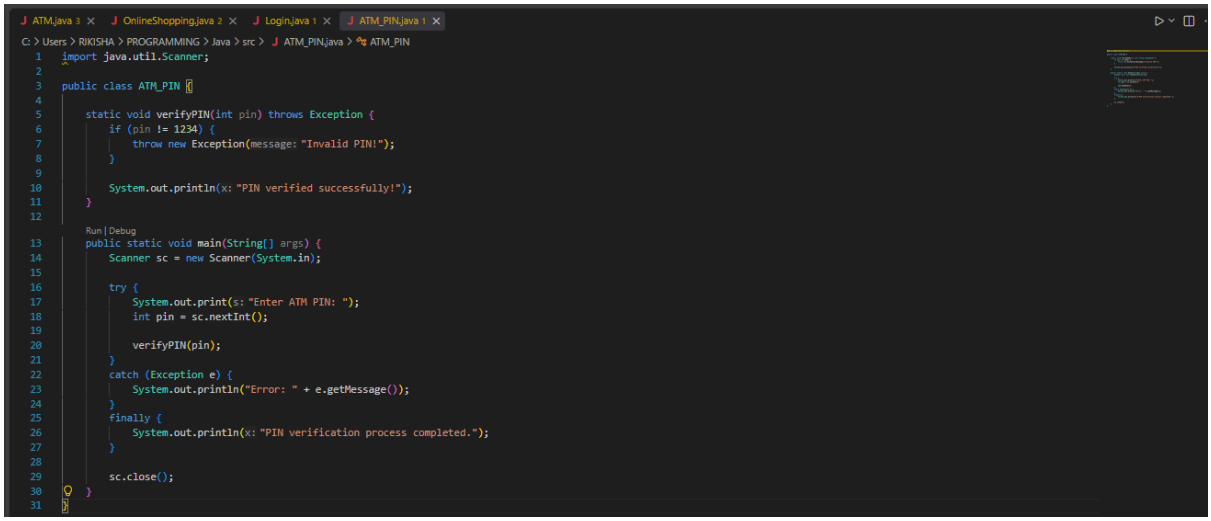
        sc.close();
    }
}
```

Output

```
PS C:\Users\RIKISHA> javac Login.java
error: file not found: Login.java
Usage: javac <options> <source files>
use --help for a list of possible options
PS C:\Users\RIKISHA> cd "C:\Users\RIKISHA\PROGRAMMING\Java\src"
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> javac Login.java
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> java Login
Enter password: riki@4321'=
Error: Invalid password!
Login process completed.
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

2. **Create an ATM PIN Verification program that throws an exception for an invalid PIN entered by the user and uses a finally block to display a message indicating that the verification process has completed.**

Programme



```java
import java.util.Scanner;

public class ATM_PIN {

    static void verifyPIN(int pin) throws Exception {
        if (pin != 1234) {
            throw new Exception(message: "Invalid PIN!");
        }

        System.out.println(x: "PIN verified successfully!");
    }

    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print(s: "Enter ATM PIN: ");
            int pin = sc.nextInt();

            verifyPIN(pin);
        }
        catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
        finally {
            System.out.println(x: "PIN verification process completed.");
        }

        sc.close();
    }
}
```

Output



```
Problems 7  Output  Debug Console  Terminal  Ports                                    powershell

PS C:\Users\RIKISHA> cd "C:\Users\RIKISHA\PROGRAMMING\Java\src"
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> javac ATM_PIN.java
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> java ATM_PIN
Enter ATM PIN: 1234
PIN verified successfully!
PIN verification process completed.
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

Conclusion

Thus, the programs successfully demonstrate exception handling in Java using throw, throws, and finally. Invalid passwords and PINs are handled appropriately, while the finally block ensures that the completion message is displayed regardless of whether an exception occurs.